vector< std::vector < smacc2::ISmaccUpdatable * > >

atomic< int64_t >

double

SmaccFifoScheduler *

processor_handle

thread

ExecutionModel

vector< smacc2::ISmaccUpdatable * >

SharedPtr

atomic< bool >

-updatableStateElements_

-lastState_

-loop_rate_hz

-scheduler_

-processorHandle_

-signalDetectorThread_

-executionModel_

-updatableClients_

-statusPub_

-end_
-initialized_
-rosInitialized_

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

map< std::string, std
::pair< std::function
< std::string()>, boost
::any > >

vector< smacc2_msgs
::msg::SmaccTransitionLogEntry >

SMRunMode

int64_t

SharedPtr

vector< smacc2::ISmaccState * >

SmaccStatus

map< std::string, std
::shared_ptr< smacc2
::ISmaccOrthogonal > >

recursive_mutex

StateMachineInternalAction

map< void *, std::shared
_ptr< smacc2::CallbackCounterSemaphore > >

#eventsLogPub_
#stateMachinePub

#stateMachineStatus
Pub_
#transitionHistory
Service_
#transitionLogPub_

SharedPtr

shared_ptr< smacc2
::introspection::SmaccState
Info >

shared_ptr< smacc2
::introspection::SmaccState
MachineInfo >

std::enable_shared
_from_this< SmaccStateInfo >

map< const std::type
_info *, std::vector
< smacc2::introspection
::ClientBehaviorInfoEntry > >

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccStateReactor
Info > > >

map< const std::type
_info *, std::vector
< std::shared_ptr< smacc2
::introspection::SmaccEventGenerator
Info > > >

int

vector< smacc2::introspection
::SmaccTransitionInfo >

vector< std::shared
_ptr< smacc2::introspection
::SmaccStateInfo > >

type_info *

string

-smaccStateMachine_

-signalDetector_

-globalData_

-transitionLogHistory_

-runMode_

-stateSeqCounter_

#timer_

#currentState_

#status_msg_

#orthogonals_

-eventQueueMutex_
-m_mutex_

-stateMachineCurrentAction_

-stateCallbackConnections_

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration
Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

#inh_

#currentStateInfo_

#stateMachineInfo_

+parentState_

+stateMachine_

+staticBehaviorInfo

+stateReactorsInfo

+eventGeneratorsInfo

+depth_
+stateIndex_

+transitions_

+children_

+id_

+demangledStateName
+fullStateName_

smacc2::introspection
::SmaccStateInfo

+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()

#node_

mutex

vector< std::shared
_ptr< smacc2::ISmaccClient > >

vector< std::vector
< std::shared_ptr< smacc2
::ISmaccClientBehavior > > >

-mutex_

#clients_

-clientBehaviors_

-stateMachine_

smacc2::ISmaccOrthogonal

+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()

boost::signals2::signal
< Signature, optional
_last_value=, typename boost
::function_traits< Signature
>::result_type >, int, std
::less< int >, function< Signature
>, typename extended_signature<
function_traits< Signature >::arity,
Signature >::function_type, boost::
signals2::mutex >

map< smacc2::ComponentKey,
std::shared_ptr< smacc2
::ISmaccComponent > >

-stateMachine_

shared_ptr< rclcpp
::Logger >

vector< std::shared
_ptr< smacc2::StateReactor > >

vector< std::shared
_ptr< smacc2::SmaccEventGenerator > >

#logger_

#stateReactors_

#eventGenerators_

#stateInfo_

smacc2::ISmaccState

+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition
TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()

-stateMachine_

#parentState_

boost::signals2::signal
< void(const WrappedResult
&), optional_last_value
< typename boost::function
_traits< void(const WrappedResult
&)>::result_type >, int, std
::less< int >, function< void(const
WrappedResult &)>, typename extended
_signature< function_traits< void(const
WrappedResult &)>::arity, void(const WrappedResult
&)>::function_type, boost::signals2::mutex >

-actionMutex_

boost::signals2::signal
< void(const Feedback
&), optional_last_value
< typename boost::function
_traits< void(const Feedback
&)>::result_type >, int, std
::less< int >, function< void
(const Feedback &)>, typename extended
_signature< function_traits< void(const
Feedback &)>::arity, void(const Feedback
&)>::function_type, boost::signals2::mutex >

smacc2::SmaccSignal
< Signature, Combiner,
Group, GroupCompare,
SlotFunction, ExtendedSlotFunction,
Mutex >

boost::signals2::signal
< void(), optional_last
_value< typename boost::
function_traits< void()>
::result_type >, int, std
::less< int >, function< void
()>, typename extended_signature
< function_traits< void()>::arity,
void()>::function_type, boost::signals2::mutex >

#components_

-orthogonal_

#stateMachine_

-goalActivityFlag_
-currentOrthogonal_

smacc2::ISmaccClient

+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()

smacc2::ISmaccClientBehavior

+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()

currentState

#actionClientName_

-goalActivityFlag_

#actionClientName_

-actionMutex_

< void(const WrappedResult &)>

< void(const Feedback &)>

< void()>

#owner_

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

optional< std::string >

optional< std::chrono
::milliseconds >

function< void(const
WrappedResult &)>

function< void(const
Feedback &)>

shared_ptr< ActionClient >

optional< std::shared
_future< typename GoalHandle
::SharedPtr > >

optional< std::shared
_future< typename rclcpp
_action::Client< ActionType
>::CancelResponse::SharedPtr > >

FeedbackCallback

smacc2::SmaccSignal
< void(const WrappedResult &)>

smacc2::SmaccSignal
< void(const Feedback &)>

smacc2::SmaccSignal
< void()>

bool

optional< std::future
< int > >

function< void()>

+actionServerName

+serverTimeout

+postAbortedEvent
+postCancelledEvent
+postSuccessEvent

+postFeedbackEvent

-client_

-lastRequest_

-lastCancelResponse_

-feedbackCallback_

+onActionAborted_
+onActionCancelled
+onActionSucceeded_

+onActionFeedback_

+onGoalAccepted_
+onGoalRejected_

-onFailure_
-onFinished_
-onSuccess_

-isShutdownRequested_

-onEntryThread_
-onExitThread_

-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn

smacc2::client_core
_components::CpActionClient
< ActionType >

+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()

smacc2::SmaccAsyncClient
Behavior

+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()

-actionClient_

+resultConnectionsInitialized_

#resultConnectionsInitialized_

ResultCode

milliseconds

#actionResult_

#goalResponseTimeout_

smacc2::client_behavior
_bases::CbActionClientBehavior
Base< TAction >

+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()

-actionClient_

#actionResult_

#goalResponseTimeout_

< nav2_msgs::action
::DriveOnHeading >

smacc2::client_behavior
_bases::CbActionClientBehavior
Base< nav2_msgs::action::DriveOnHeading >

+ onStateOrthogonalAllocation()
+ ~CbActionClientBehaviorBase()
+ onExit()
# sendGoal()
# cancelGoal()
# onActionSuccess()
# onActionAbort()
# onActionFeedback()
# setActionClientName()
- dispatchActionSuccess()
- dispatchActionAbort()
- dispatchActionFeedback()

#actionClientName_

float

seconds

-distance_
-speed_

-timeAllowance_

cl_nav2z::CbDriveOnHeading

+ CbDriveOnHeading()
+ onEntry()